

Oklahoma's Buried Meteor Crater Is 100 Million Years Younger Than Thought

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Deep beneath the small town of Ames, Oklahoma, lies a giant meteor crater that stretches for miles underground. Layers of **sediment** now cover it, but scientists have just learned something surprising about it. The crater is nearly 100 million years younger than they believed.

For years, researchers thought the Ames crater formed about 467.5 million years ago. That would have made it part of a group of big meteor impacts across North America known as the Ordovician Meteor Event. Because so many craters seemed to date from the same time, some scientists have suggested Earth may once have had a ring of space rocks around it, a bit like Saturn's rings.

Now a team from The University of Texas at Austin has a new answer. They studied tiny crystals called **zircons**, found in granite rock that was changed by the impact. By dating those crystals, they found the meteor struck about 370 million years ago instead. "No matter what technique we used, it was coming back to this younger signal," said lead author Elizabeth Catlos.

So why was the old date wrong? Earlier, scientists had dated the crater using fossil teeth from an ancient eel-like animal called a **conodont**. Catlos said those teeth were probably already millions of years old when the meteor hit. The impact likely churned up older material and mixed the fossils into the rocks.

The team worked with NASA to image the **zircons** and confirm they were changed by the crash. The new date places the impact close to a **mass extinction** about 372 million years ago that wiped out much of the ocean's life. The study was published in July in the journal *Meteoritics & Planetary Science*.

Word Treasure

impact crater -- A hole formed when a space rock strikes a planet.

zircon -- A durable mineral crystal used to date rocks.

conodont -- An ancient eel-like animal known from its fossil teeth.

mass extinction -- A time when a large share of living things died out.

sediment -- Layers of sand, mud and other material that settle over time.

Break it down

Lead: a buried Oklahoma crater is much younger than thought

The old theory and the possible ring of debris

Zircon dating gives a new age

Why fossil teeth gave a misleading date

Confirmation with NASA and the link to a mass extinction

Why it matters

Correct dates for ancient impacts help scientists test ideas about Earth's history, including whether impacts are linked to mass extinctions.

Test what you remember

1. About when did scientists previously think the Ames crater formed?
 - (A) 370 million years ago
 - (B) 467.5 million years ago
 - (C) 10,000 years ago
 - (D) 1 billion years ago
2. What is the group of impacts the crater was thought to belong to called?
 - (A) The Ordovician Meteor Event
 - (B) The Ice Age Impact
 - (C) The Devonian Flood
 - (D) The Saturn Ring Event
3. Which university led the new study?
 - (A) The University of Melbourne
 - (B) The University of Texas at Austin
 - (C) Johns Hopkins University
 - (D) University of Wisconsin-Madison
4. What did Elizabeth Catlos say kept happening no matter which technique they used?
 - (A) The crater disappeared
 - (B) They found dinosaur bones
 - (C) The crystals broke
 - (D) It kept coming back to a younger signal
5. Which agency helped image the zircons?
 - (A) NOAA
 - (B) NASA
 - (C) The FBI
 - (D) ESA
6. What event is the new date close to?
 - (A) The start of the Ice Age
 - (B) The first Moon landing
 - (C) The formation of Saturn's rings
 - (D) A mass extinction about 372 million years ago

Answer key (try the quiz first!): 1: B 2: A 3: B 4: D 5: B 6: D

MY THOUGHTS (at least 20 words):